

R2 — ML Research Track, Week 1

Member: _____

Project setup (do first)

1. Fork the repo: <https://github.com/slvDev/esp32-ai>
2. Clone your fork. Add the original as upstream.
3. Read README.md, RESULTS.md, and research/tinystories/README.md.
4. Write 5 lines in the shared Drive doc: what you think the project does and where the hard part is.

Your task this week

Goal: one sheet holds every published claim and its status.

Steps: 1. Read RESULTS.md again. List every number in it. 2. Build a Google Sheet. Columns: claim, published value, our value, status, owner. 3. Add one row per claim. Status is "pending" for all rows. 4. Share the sheet with the club Drive folder.

Deliverable: the sheet link in the club Drive, with at least 20 claim rows.

Verify: R1 finds the data claim in the sheet without help.

Learning this week

Neural network basics. Watch episodes 4 and 5 of 3Blue1Brown (backpropagation). Write the XOR MLP with one hidden layer and backprop.

Resources

- RESULTS.md: <https://github.com/slvDev/esp32-ai/blob/main/RESULTS.md>
- 3Blue1Brown neural networks: https://www.youtube.com/playlist?list=PLZHQObOWTQDNU6R1_67000Dx_ZCJB-3pi
- micrograd (minimal autograd): <https://github.com/karpathy/micrograd>

Done when

- [] Tracking sheet has at least 20 claim rows, all pending
- [] Sheet link is in the club Drive
- [] XOR MLP with backprop runs
- [] You can say what the project does in one sentence